

## **Math 104 – Real Analysis**

Homework 2 — my solutions

Fall 2026   ■   Keshav Ramamurthy

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## Homework 2

Deadline: 11:59 pm September 13, 2026

**Problem (1).** Let  $S$  be a nonempty bounded subset of  $\mathbb{R}$ , and define the diameter of  $S$  by

$$\text{diam}(S) = \sup\{|s - t| : s, t \in S\}.$$

Prove that  $\text{diam}(S) = \sup S - \inf S$ .

**Problem (2).** Prove that for every real number  $a > 0$  there is a unique natural number  $n$  such that

$$n - 1 \leq a < n.$$

(Your proof will need both the Archimedean Property and the Well-Ordering Principle for  $\mathbb{N}$ , together with an argument for uniqueness.)

**Problem (3).** Let

$$S = \{x \in \mathbb{R} : x^4 - 5x^2 + 4 < 0\}.$$

(a) Write  $S$  in interval notation.

(b) Find  $\inf S$  and  $\sup S$ , and state clearly whether each is attained.

**Problem (4).** For each sequence below, determine whether it converges and, if so, find its limit. (You may use the limit theorems; no  $\varepsilon$ - $N$  proof is required here, but you must justify divergence when it occurs, e.g. by exhibiting two subsequences with different limits.)

$$(a) a_n = \frac{2n^3 + n}{n^3 - 5}, \quad (b) b_n = (-1)^n \frac{n}{n + 1}.$$

**Problem (5).** Give rigorous  $\varepsilon$ - $N$  proofs of the following limit.

$$\lim_{n \rightarrow \infty} \frac{3n^2 - 1}{2n^2 + n + 5} = \frac{3}{2}.$$

## Solutions

### Problem 1

*Solution.*

□

### Problem 2

*Solution.*

□

### Problem 3

*Solution.*

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### Problem 4

*Solution.*

□

### Problem 5

*Solution.*

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